

CBSE Class 10 Mathematics (Standard) 2019

Set 1 | Q.P. Code 30/1/1

SECTION A

Multiple Choice Questions (1 Mark Each)

Q1. If α and β are the zeroes of the polynomial $(3x^2 + 6x + k)$ such that $(\alpha + \beta + \alpha\beta = -\frac{2}{3})$, then find the value of (k) .

Q2. If $(x = 1)$ and $(y = 2)$ is a solution of the pair of linear equations $(2x - 3y + a = 0)$ and $(2x + 3y - b = 0)$, then:

Q3. The midpoint of the line segment joining the points $(P(-4,5))$ and $(Q(4,6))$ lies on:

Q4. If θ is an acute angle and $(7 + 4\sin\theta = 9)$, then find θ .

Q5. Find the value of:

$$\left[\tan^2\theta - \left(\frac{1}{\cos\theta} \right) \times \sec\theta \right]$$

Q6. If $HCF(98, 28) = m$ and $LCM(98, 28) = n$, find $(n - 7m)$.

Q7. The tangents drawn at the extremities of a diameter of a circle are always:

Q8. In triangles ABC and DEF , if $\angle B = \angle E$, $\angle F = \angle C$ and $AB = 3DE$, then the two triangles are:

Q9. If

$$\left[(-1)^n + (-1)^8 = 0 \right]$$

then (n) is:

Q10. Two polynomials are shown in the graph. Find the number of distinct zeroes of both polynomials together.

Q11. If the sum of first (m) terms of an AP is

$$\begin{bmatrix} 2m^2 + 3m \\ \end{bmatrix}$$

find its second term.

Q12. Mode and Mean of a data are respectively $15x$ and $18x$. Find the Median.

Q13. A card is selected at random from a deck of 52 cards. Find the probability that it is a red face card.

Q14. Which of the following is a rational number between $(\sqrt{3})$ and $(\sqrt{5})$?

Q15. If the area of a sector is (40π) square units and the angle at the centre is (72°) , find the radius of the circle.

Q16. In the given figure, PA is a tangent from an external point P to a circle with centre O. If $\angle POB = 115^\circ$, find $\angle APO$.

Q17. A kite is flying at a height of 150 m from the ground. Its string makes an angle of (30°) with the horizontal. Find the length of the string.

Q18. A wire 20 cm long is bent into the form of an arc of a circle of radius $(\frac{60}{\pi})$ cm. Find the angle subtended by the arc at the centre.

Assertion–Reason Questions

Q19.

Assertion (A): The probability of selecting a number at random from the numbers 1 to 20 is 1.

Reason (R): For any event E, if $(P(E)=1)$, then E is called a sure event.

Q20.

Assertion (A): If two hemispheres of the same radius are joined along their bases, a sphere is obtained.

Reason (R): Total surface area of a sphere of radius (r) is $(3\pi r^2)$.

SECTION B

Very Short Answer Questions (2 Marks Each)

Q21. (a) If

$$[\begin{aligned} &x\cos 60^\circ + y\cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5 \\ &] \end{aligned}$$

find the value of $(x+2y)$.

OR

(b) Evaluate:

$$[\begin{aligned} &\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ} \\ &] \end{aligned}$$

Q22. Find the zeroes of the polynomial

$$[\begin{aligned} &p(x) = x^2 + \frac{4}{3}x - \frac{4}{3} \\ &] \end{aligned}$$

Q23. The coordinates of the centre of a circle are $((2a, a-7))$. If the circle passes through the point $((11, -9))$ and has diameter $(10\sqrt{2})$ units, find the value of (a) .

Q24. (a) If

$$[\begin{aligned} &\triangle ABC \sim \triangle PQR \\ &] \end{aligned}$$

where

$AB = 6$ cm, $BC = 4$ cm, $AC = 8$ cm and $PR = 6$ cm,

find the length of $((PQ + QR))$.

OR

(b) In the given figure, if

$$[\begin{aligned} &\frac{QR}{QS} = \frac{QT}{PR} \\ &] \end{aligned}$$

and $\angle 1 = \angle 2$, show that

[
\triangle PQS \sim \triangle TQR
]

Q25. A person stands at a point P outside a circular ground at a distance of 26 m from its centre. The lengths of tangents PA and PB drawn from P are 10 m each. Find the radius of the circular ground.